HEART DISEASE CLASSIFICATION

SCREENSHOTS

[SUCCESSFUL API HEALTH CHECK]

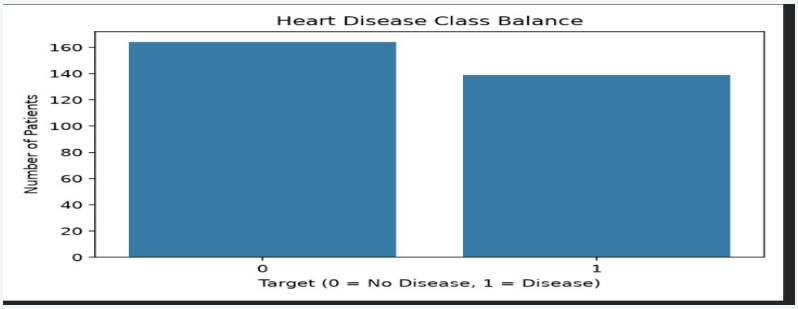
Pretty-print

{"status":"ok","model_loaded":true}

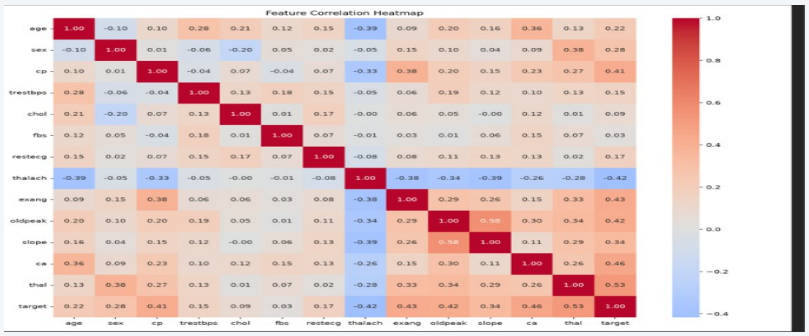
[SCREENSHOT : DATA PREPARATION / DATASET PREVIEW]

```
(heart-mlops) C:\Users\Prakash\heart-disease-mlops>python -c "import pandas as pd; d=pd.read_csv('data/heart.csv'); print(d.head()); print(d.shape); print(d
.isna().sum()); print(d['target'].value_counts())"
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope   ca  thal  target
0   63    1   1       145   233    1        2      150      0      2.3      3  0.0   6.0       0
1   67    1   4       160   286    0        2      108      1      1.5      2  3.0   3.0       1
2   67    1   4       120   229    0        2      129      1      2.6      2  2.0   7.0       1
3   37    1   3       130   250    0        0      187      0      3.5      3  0.0   3.0       0
4   41    0   2       130   204    0        2      172      0      1.4      1  0.0   3.0       0
(303, 14)
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          4
thal        2
target      0
dtype: int64
target
0    164
1    139
Name: count, dtype: int64
```

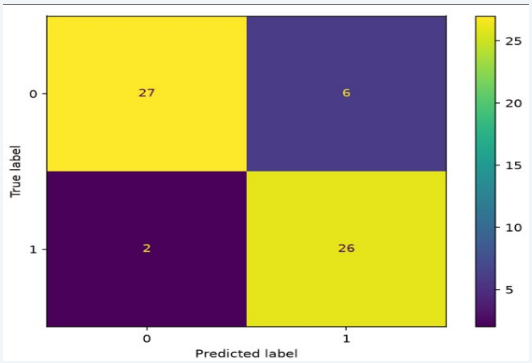
[EDA FIGURE 1 — TARGET CLASS BALANCE]



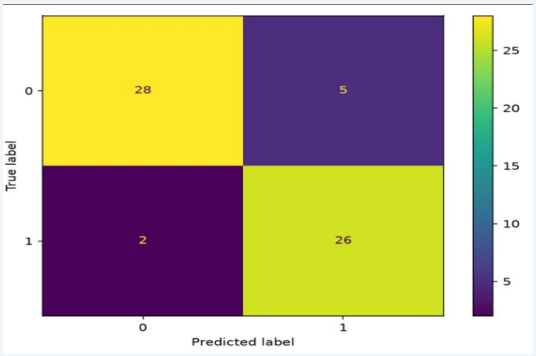
[CORRELATION HEATMAP]



[TRAINING OUTPUT / MODEL COMPARISON]



Logistic Regression

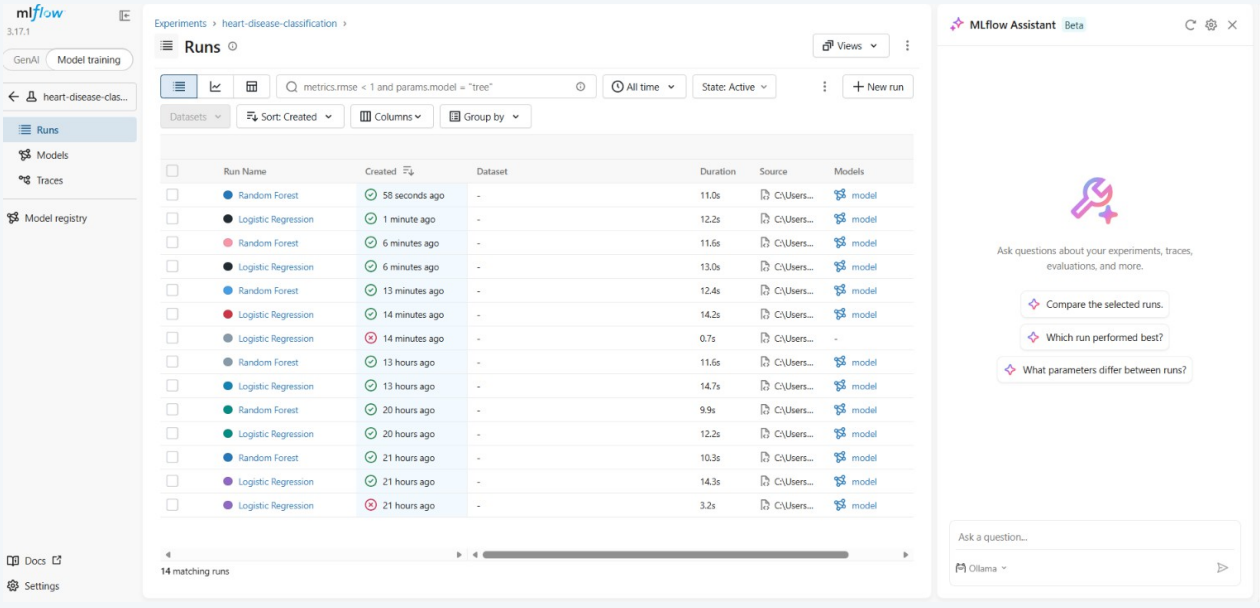


Random Forest

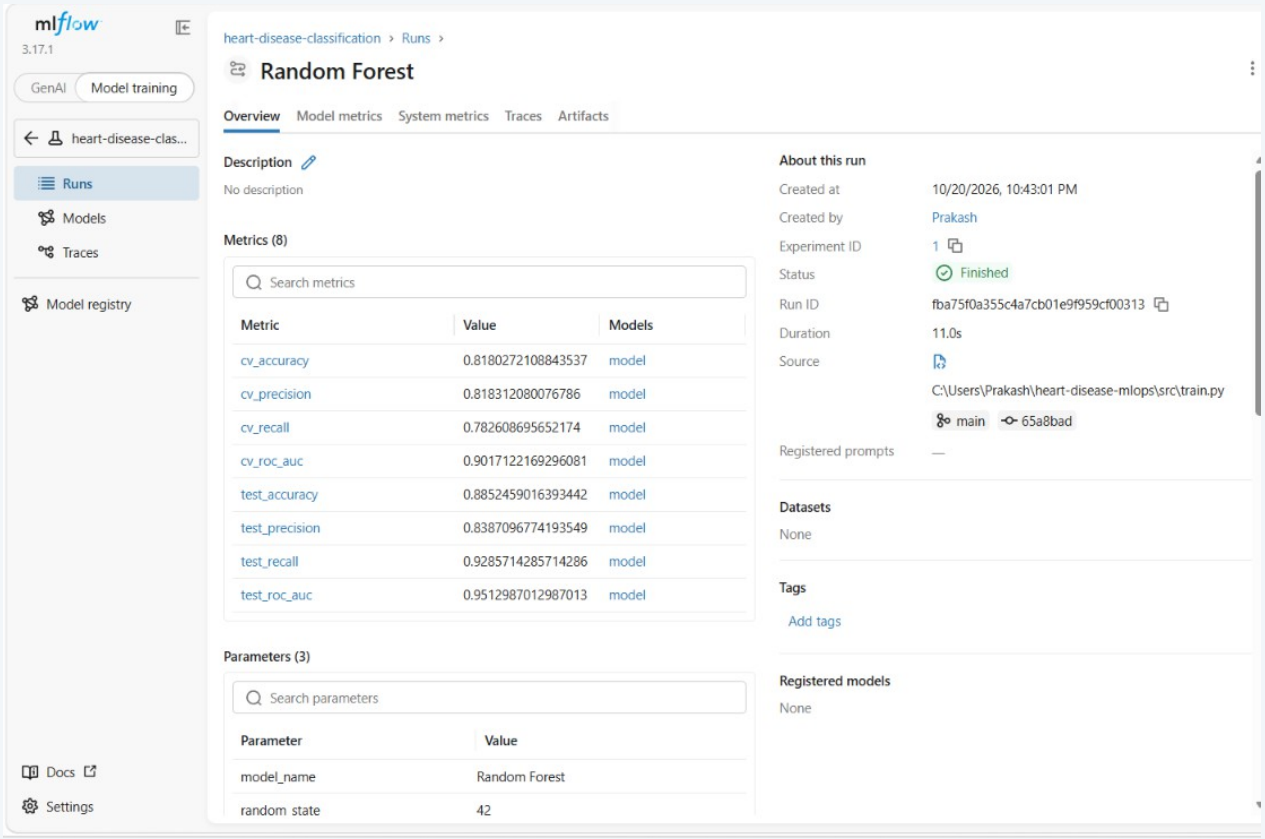
```
Results for Logistic Regression
{
  "cv_accuracy": 0.8551870748299321,
  "cv_precision": 0.8730744935655421,
  "cv_recall": 0.8098814229249012,
  "cv_roc_auc": 0.9024779568257829,
  "test_accuracy": 0.8688524590163934,
  "test_precision": 0.8125,
  "test_recall": 0.9285714285714286,
  "test_roc_auc": 0.9664502164502166
}
MLflow run ID: c47c5ff16d97428ba36de8110cb4f317
2026/10/20 22:30:16 WARNING mlflow.sklearn: Saving scikit-learn models in the pickle or cloudpickle format requires exercising caution because these formats
 rely on Python's object serialization mechanism, which can execute arbitrary code during deserialization. The recommended safe alternative is the 'skops' f
ormat. For more information, see: https://scikit-learn.org/stable/model_persistence.html

Results for Random Forest
{
  "cv_accuracy": 0.8180272108843537,
  "cv_precision": 0.8183120880076786,
  "cv_recall": 0.782608695652174,
  "cv_roc_auc": 0.9017122169296081,
  "test_accuracy": 0.8852459016393442,
  "test_precision": 0.8387096774193549,
  "test_recall": 0.9285714285714286,
  "test_roc_auc": 0.9512987012987013
}
MLflow run ID: df8e7307105f45c2a83f541d5bc84cd1
```

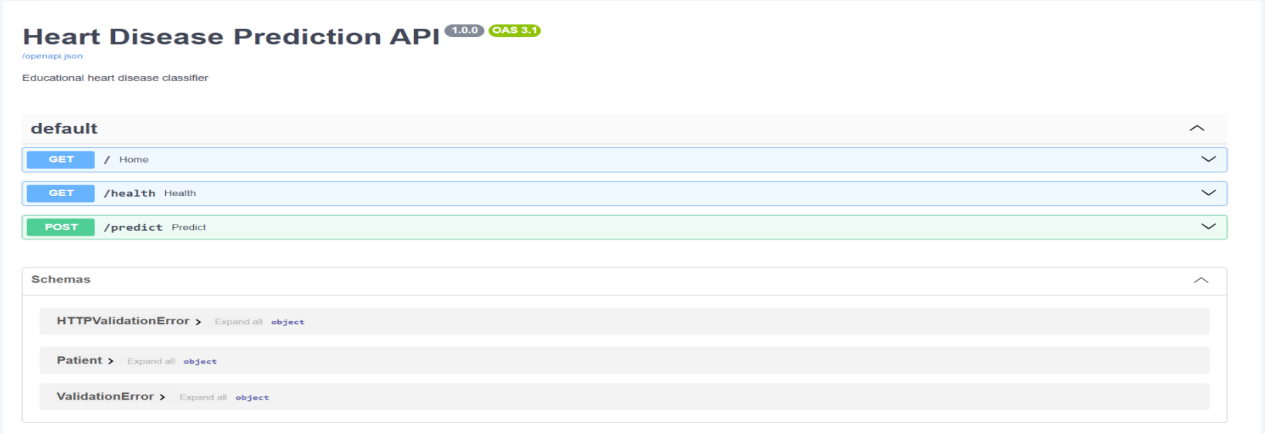
[MLFLOW EXPERIMENT OVERVIEW]



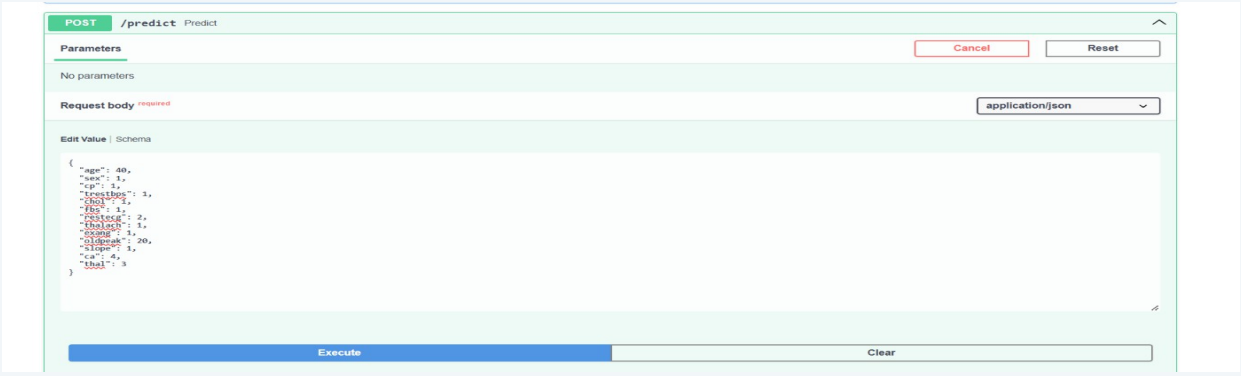
[MLFLOW RUN DETAILS]



[FASTAPI SWAGGER UI]



[POST: PREDICTION RESPONSE]

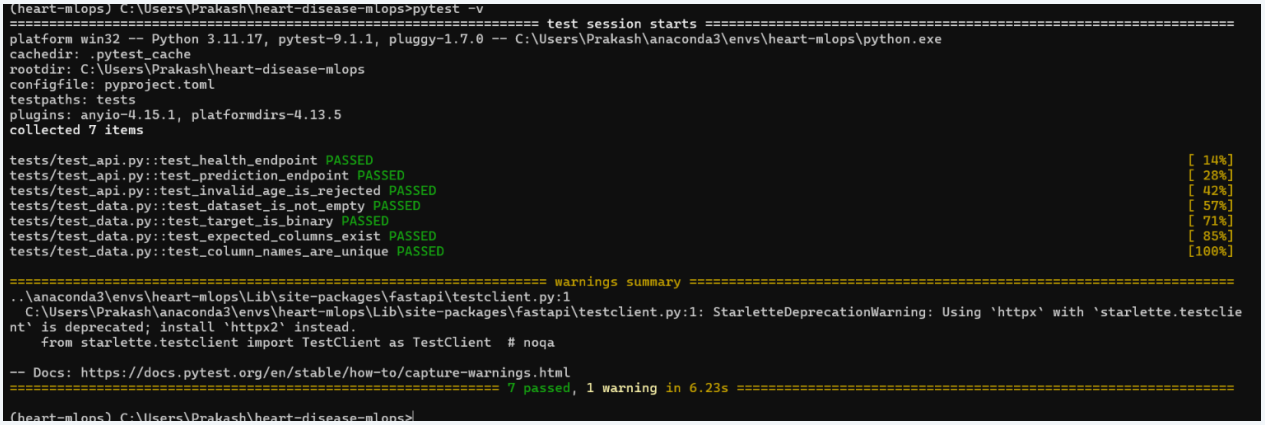


[PREDICTION RESPONSE]



t

[GITHUB ACTIONS SUCCESSFUL RUN]



[DOCKER BUILD / CONTAINER HEALTH]

```
(heart-mlops) C:\Users\Prakash\heart-disease-mlops>docker build -t heart-disease-api:1.0 .
[+] Building 2.9s (12/12) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                              0.1s
 => => transferring dockerfile: 334B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                               2.3s
 => [auth] library/python:pull token for registry-1.docker.io                                                                     0.0s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 159B                                                                                                 0.0s
 => [1/6] FROM docker.io/library/python:3.11-slim@sha256:e88e9763f943ec1834f992a4b51e0f24500486803e8bc534e5767af9ea65f6ce         0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:e88e9763f943ec1834f992a4b51e0f24500486803e8bc534e5767af9ea65f6ce         0.1s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 6.33kB                                                                                               0.0s
 => CACHED [2/6] WORKDIR /app                                                                                                     0.0s
 => CACHED [3/6] COPY requirements.txt .                                                                                          0.0s
 => CACHED [4/6] RUN pip install --no-cache-dir -r requirements.txt                                                               0.0s
 => CACHED [5/6] COPY src/ ./src/                                                                                                 0.0s
 => CACHED [6/6] COPY models/ ./models/                                                                                           0.0s
 => exporting to image                                                                                                            0.2s
 => => exporting layers                                                                                                           0.0s
 => => exporting manifest sha256:0697504b9fea46a34569e01358e6d3a83fa95ca61e88dff9ce89eef373f30620                                 0.0s
 => => exporting config sha256:bc715dc291875a4b38cdf32b28763104fffc66d6fe2430a0184fc670be00a1aa                                   0.0s
 => => exporting attestation manifest sha256:00a5ba945eefd29aacaf44805cfea6ae876abbaab4b2e83346179caa3bf0ecee                     0.0s
 => => exporting manifest list sha256:06e6f4b3812d16d924e6935cbdfe854fdf43a46f9ae148034c258b054f301a5a                            0.0s
 => => naming to docker.io/library/heart-disease-api:1.0                                                                          0.0s
 => => unpacking to docker.io/library/heart-disease-api:1.0                                                                       0.0s

(heart-mlops) C:\Users\Prakash\heart-disease-mlops>docker run --rm -p 8000:8000 heart-disease-api:1.0
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     172.17.0.1:42304 - "GET /health HTTP/1.1" 200 OK
```

[SCREENSHOT : KUBERNETES RESOURCES]

```
Anaconda Prompt

(base) C:\Users\Prakash>curl.exe http://localhost:8000/health
{"status":"ok","model_loaded":true}
(base) C:\Users\Prakash>
```

[SCREENSHOT : FINAL DEPLOYED API EVIDENCE]

```
(base) C:\Users\Prakash>kubectl describe deployment heart-disease-api
Name:                   heart-disease-api
Namespace:              default
CreationTimestamp:      Tue, 20 Oct 2026 14:47:18 +0530
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=heart-disease-api
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=heart-disease-api
  Containers:
   api:
    Image:      heart-disease-api:1.0
    Port:       8000/TCP
    Host Port:  0/TCP
    Limits:
      cpu:     500m
      memory:  512Mi
    Requests:
      cpu:         100m
      memory:      128Mi
    Liveness:      http-get http://:8000/health delay=10s timeout=1s period=20s #success=1 #failure=3
    Readiness:     http-get http://:8000/health delay=5s timeout=1s period=10s #success=1 #failure=3
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   heart-disease-api-7f6985775b (1/1 replicas created)
Events:          <none>

(base) C:\Users\Prakash>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS GATES
heart-disease-api-7f6985775b-qz7qb   1/1     Running   0          8h    10.244.0.3   minikube   <none>           <none>

(base) C:\Users\Prakash>
```